

RAS Water Quality Summary - January 2026

Bayshore Sustainable Aquaculture | Grow-out Hall A (L. vannamei) | Prepared by A. Cross

Daily morning readings taken at 05:00 CT from the YSI ProDSS handheld, port G-2, three-tank composite. Values logged before first feed. This summary rolls up the daily field log for the Lonestar Foundation quarterly file and the internal husbandry record.

Control targets

Parameter	Target	Unit
Temperature	28.0 - 30.0	C
Dissolved oxygen	>= 5.0	mg/L
pH	7.6 - 8.2	-
Salinity	16 - 22	ppt
Total ammonia nitrogen	< 0.50	mg/L
Nitrite (NO ₂ -N)	< 1.00	mg/L
Nitrate (NO ₃ -N)	< 60	mg/L
Alkalinity (CaCO ₃)	120 - 160	mg/L

Daily readings (first half of month)

Date	Temp C	DO mg/L	pH	Sal ppt	TAN mg/L	NO ₂ mg/L
2026-01-01	28.6	6.0	8.16	19.5	0.30	0.41
2026-01-02	28.8	5.6	8.08	20.0	0.23	0.24
2026-01-03	29.6	6.2	7.99	18.6	0.32	0.22
2026-01-04	29.7	5.8	7.67	17.8	0.10	0.56
2026-01-05	29.3	6.2	8.14	19.1	0.33	0.39
2026-01-06	29.4	6.0	7.96	20.5	0.26	0.26
2026-01-07	29.5	6.4	7.65	19.8	0.31	0.18
2026-01-08	29.4	6.6	8.10	18.4	0.23	0.60
2026-01-09	29.6	6.3	8.12	18.2	0.22	0.37
2026-01-10	28.5	5.7	7.75	18.3	0.26	0.22
2026-01-11	29.0	5.6	7.67	20.0	0.33	0.43
2026-01-12	28.8	7.2	7.91	18.2	0.27	0.49
2026-01-13	29.3	5.6	7.84	20.0	0.08	0.59
2026-01-14	28.7	6.9	7.68	20.2	0.36	0.41

Notes and actions

- Jan 6: TAN crept to 0.41 mg/L on Tank 2 after a heavy feed day. Cut ration 15 percent for 48 hours, increased biofilter backwash; back under 0.25 by Jan 8.

- Jan 11: salinity drift up to 21.0 ppt from evaporation on the chiller loop. Topped with dechlorinated freshwater to bring the hall back to 18.5 ppt.
- Jan 14: dissolved oxygen held above 5.4 mg/L through a mild spell; backup blower test ran clean.
- No nitrite excursions above 1.0 mg/L this period. Alkalinity dosed twice with sodium bicarbonate to hold the 120-160 band.

Overall the hall ran inside spec for the month. The only watch item is afternoon temperature on warm days; recommend pulling a second reading at 14:00 once the cycle biomass climbs.

Internal husbandry record. Figures feed the quarterly grant report and the harvest forecast. Retain 3 years.